

Ethical Hacking Task 02

Network Scanning & Service Enumeration Report

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Target System	localhost (127.0.0.1)
Tool Used	Nmap
Task	Ethical Hacking Task 02

Part A – Nmap Installation

Nmap was successfully installed and verified using the command 'nmap --version'. The installation output was captured as evidence.

Part B – Local Machine Scan

A scan of localhost (127.0.0.1) was performed to identify active open ports and running services. The results were documented using screenshots.

Part C – Service Version Detection

The command 'nmap -sV localhost' was executed to identify service names and software versions running on detected ports.

Part D – Operating System Detection

Operating system detection was performed using 'nmap -O localhost'. This technique helps identify the target operating system for security assessments.

Part E – Common Port Research

Port	Protocol	Purpose	Risk
20/21	FTP	File Transfer	Weak passwords
22	SSH	Secure Remote Login	Brute-force attacks
23	Telnet	Remote Login	Unencrypted traffic
25	SMTP	Email	Spam/Spoofing
53	DNS	Name Resolution	DNS spoofing
80	HTTP	Web	Plain-text traffic
110	POP3	Email	Credential theft
143	IMAP	Email	Misconfiguration
443	HTTPS	Secure Web	TLS issues

445	SMB	File Sharing	SMB exploits
3389	RDP	Remote Desktop	Brute-force attacks

Part F – Scan Analysis

Only required services should remain enabled. Services such as FTP, Telnet, SMB and RDP may become security risks if left exposed or improperly configured. If Telnet is not required, it should be disabled because it transmits data without encryption.

Observations

The scan successfully identified network information from the authorized local system. Nmap proved effective for discovering services and gathering basic security information while using safe scanning techniques.

Conclusion

This task provided practical experience with network scanning and service enumeration using Nmap. By scanning the local machine, it was possible to identify open ports, running services, service versions, and operating system information. Understanding common ports and the risks associated with exposed services is an important part of ethical hacking. The activity also reinforced the importance of scanning only systems that you own or have permission to test. Proper network scanning helps security professionals reduce attack surfaces, identify unnecessary services, and improve the overall security posture of a system.